

 YTU – Faculty of Chemical and Metallurgical Engineering, Questions and Answers Sheet		NOTE CHART					
							Total
Student Name and Surname							
Students Number							
Section						Exam Date	16 / 04 / 201
Course Name	Introduction to Financial Mathematics	Group Number	3 - 4	Exam Duration		Examination Room	
Course Instructor Name and Surname	Assist. Prof. Dr. Hale Gonce Kocken				Signature		
Student Disciplinary Regulations "and to make or attempt to make copies of exams to" the actual perpetrators are suspended from one or two semesters. (YÖK; 2547 Student Disciplinary Regulations, 9. Article)							

- (25) Q1) Consider an initial investment of \$100 using an amount function which is a second degree polynomial. You are given that the future values at $t = 1$ and $t = 2$ are 120 and 158, respectively.
- Determine i_3 .
 - Determine the equivalent effective rate of discount between $t = 0.5$ and $t = 1.5$.
 - Determine $\delta_{1,2}$.
 - Find the present value of 1 to be paid at time $t = 1.25$.
- (10) Q2) Suppose that you have \$20000 cash today and can invest it at an effective interest rate of 10%. How many years will it take you to become a millionaire?
- (15) Q3) Bruce deposits 100 into a bank account. His account is credited interest at an annual nominal rate of interest of %4 convertible semiannually. At the same time, Peter deposits 100 into a separate account. Peter's account is credited interest at an annual force of interest of δ . After 7 years and 3 months, the value of each account is the same. Calculate δ .
- (20) Q4) Jeff deposits 10 into a fund today and 20 fifteen years later. Interest for the first ten years is credited at a nominal discount rate of d compounded quarterly, and thereafter at a nominal interest rate of 6% compounded semiannually. The accumulated balance in the fund at the end of 30 years is 100. Calculate d .
- (15) Q5) On January 1, 2006, Jack deposited 1,000 into Bank X to earn interest at the rate of j per annum compounded semi-annually. On January 1, 2011, he transferred his account to Bank Y to earn interest at the rate of k per annum compounded quarterly. On January 1, 2014, the balance at Bank Y is 1,990.76. If Jack could have earned interest at the rate of k per annum compounded quarterly from January 1, 2006 through January 1, 2014, his balance would have been 2,203.76. Calculate the ratio k/j .
- (15) Q6) The amount of interest earned on A for one year is \$150, while the equivalent amount of discount is \$210. Find A.
- 150 210

Q1) $A(t) = at^2 + bt + c$

$A(0) = 100$ $A(1) = 120$ $A(2) = 158$

$c = 100$ $A(1) = a + b + c = 120$

5x5

$a + b = 20$

$A(2) = 4a + 2b + c = 158$

$4a + 2b = 58$

25

$2a = 18$

$a = 9$

$b = 11$

$A(t) = 9t^2 + 11t + 100$

a) $i_3 = \frac{A(3) - A(2)}{A(2)} = \frac{9 \cdot 9 + 11 \cdot 3 + 100 - 158}{158} = \frac{56}{158} = 0.35$

b) $d = \frac{A(1.5) - A(0.5)}{A(1.5)} = \frac{136.75 - 107.75}{136.75} = 0.21$

c) $\delta = \frac{A'(t)}{A(t)} = \frac{18t + 11}{9t^2 + 11t + 100} \Rightarrow \delta_{1.2} = \frac{32.6}{126.16} = 0.2584 = 0.26$

d) $\begin{array}{cc} X \swarrow & \searrow 1 \\ | & | \\ t=0 & t=1.25 \\ A(0) & A(1.25) \\ =100 & =127.81 \end{array}$

$\frac{A(0)}{A(1.25)} = \frac{x}{1}$

$x = 0.78$

Q2) $\begin{array}{cc} \$20000 & \$1,000,000 \\ | & | \\ t=0 & t=n \end{array}$

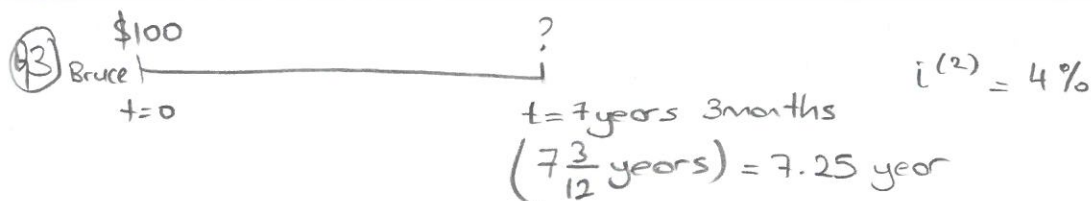
$20000(1+i)^n = 1000000$

$(1+0.1)^n = 50$

$n \times \ln 1.1 = \ln 50$

$n = 41.045$

10



$$FV_{\text{Bruce}} = 100 \cdot \left(1 + \frac{0.04}{2}\right)^{2 \times (7.25)} = 133.2608$$

87 months

$$7.25 = \frac{29}{4} \text{ years.}$$

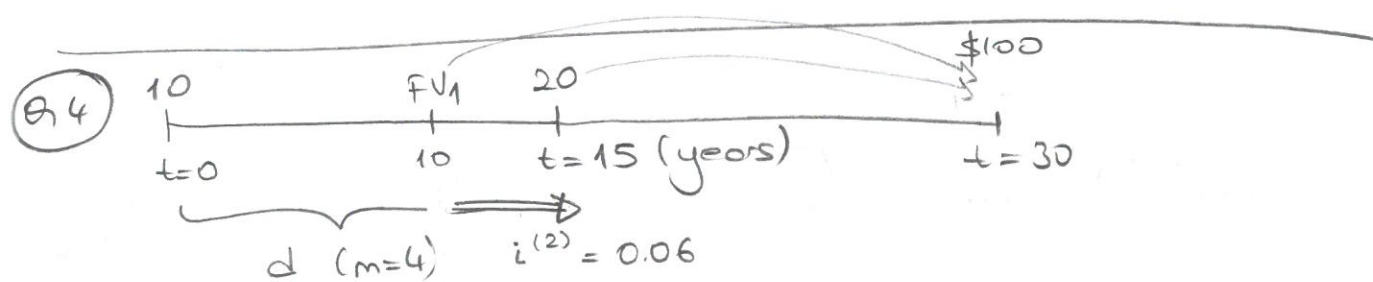


$$FV_{\text{Peter}} = 100 \cdot e^{8 \cdot (7.25)} = 133.2608$$

$$\ln e^{8 \cdot (7.25)} = \ln(1.3326)$$

$$8(7.25) = 0.2871$$

$$\boxed{8 = 0.0396}$$



$$FV_1 = 10 \cdot \left(1 - \frac{d^{(4)}}{4}\right)^{-4 \times 10} =$$

$$FV_1 \cdot \left(1 + \frac{i^{(2)}}{2}\right)^{2 \times 20} + 20 \cdot \left(1 + \frac{i^{(2)}}{2}\right)^{2 \times 15} = 100 \quad \text{at } i^{(2)} = 0.06$$

48.5452

$$FV_1 (1.03)^{40} = 51.4548$$

$$FV_1 = 15.7738$$

$$10 \left(1 - \frac{d}{4}\right)^{-40} = 15.7738$$

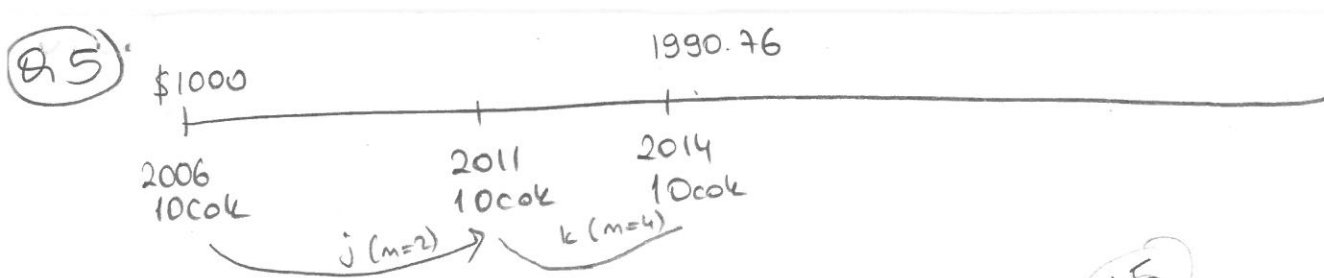
$$\frac{d}{4} = 0.011329$$

$$\boxed{d = 0.0453178}$$

$$\left(1 - \frac{d}{4}\right)^{-40} = 1.57738$$

$$1 - \frac{d}{4} = (1.57738)^{-1/40}$$

$$d = 4(1 - (1.57738)^{-1/40})$$



$$1000 \left(1 + \frac{j}{2}\right)^{2 \times 5} \cdot \underbrace{\left(1 + \frac{k}{4}\right)^{4 \times 3}}_{1.344888} = 1990.76$$

$$1000 \left(1 + \frac{k}{4}\right)^{4 \times 8} = 2203.76$$

$$\frac{k}{j} = ?$$

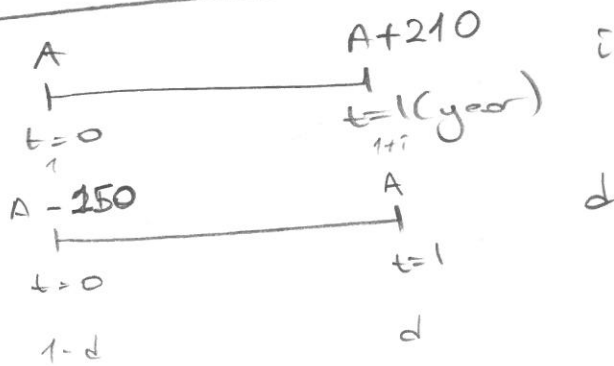
$$\boxed{k = 0.1}$$

$$\Rightarrow \frac{k}{j} = \frac{0.1}{0.08} = 1.25$$

$$j = 0.07999$$

$$\cong 0.08 //$$

Q6



$$A(1-d) = A - 150$$

$$A - Ad = A - 150$$

$$A \cdot i = 210$$

$$A \cdot d = 150$$

$$d = \frac{i}{i+1}$$

$$\frac{150}{A} = \frac{\frac{210}{A}}{1 + \frac{210}{A}} = \frac{210}{A+210}$$

$$150A + 150 \times 210 = 210 \times A$$

$$150 \times 210 = 60 \times A$$

$$\boxed{A = 525}$$

